**QP CODE: 25901708**

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE
EXAMINATION APRIL 2026

SECOND SEMESTER
Discipline Specific Core Course (DSC) - MG2DSCHTC100 - PLANTPROPAGATION

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes Max. Marks : 50

***Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I)and Appreciation(Ap)***

Students should attempt at least one question from each course outcome to enhance their overalloutcome attainability.

**Part A -** Short Answer - Answer any **10** questions - Each question carries **1** marks

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| 1 | Name any four tools used in plant propagation. | [K] / [CO1] |
| | Secateurs/pruning shears (for cutting stems), grafting knife, and budding knife.<br>Hand trowel (digging/planting), dibber (making holes for seeds/cuttings), and potting fork**.** ***(any four tools)*** | |
| 2 | Write any two types of artificial method of plant propagation. | [K] / [CO2] |
| | Cutting, grafting, budding, layering, micro propagation (any two) | |
| 3 | Compare and contrast the functions of the transfer area and incubation area in a tissue culture lab. | [U] / [CO2] |
| | Transfer area (or inoculation room) - is a dedicated, sterile zone primarily used for aseptically manipulating plant or animal tissues to prevent contamination<br>Incubation area - is a dedicated, controlled environment designed to promote the growth, differentiation, and maintenance of cultured cells, tissues, or plantlets (explants). | |
| 4 | Discuss the role of organic supplements in a tissue culture medium. | [U] / [CO2] |
| | Organic supplements in tissue culture media - including vitamins, amino acids, and complex extracts like coconut water - act as vital growth enhancers, stimulating accelerated cell proliferation, shoot regeneration, and vigorous root development. | |
| 5 | Define Ring budding. | [K] / [CO3 |
| | Ring budding, also known as annular budding, is a specialized plant propagation technique where a complete ring of bark (carrying a bud) is removed from a scion branch and fitted onto a similarly prepared, de-topped rootstock of the same diameter. | |
| 6 | What is a phytotron? | [U] / [CO4] |
| | A phytotron is a specialized research facility or enclosed laboratory designed to study plant growth and interactions with the environment by providing precisely controlled conditions | |
| 7 | Why are artificial propagation methods used in agriculture? | [A] / [CO3] |
| | Artificial propagation methods—such as cuttings, grafting, and tissue culture—are used in agriculture to rapidly produce large numbers of genetically identical, high-quality plants | |
| 8 | What is callus culture? | [K] / [CO3] |
| | Callus culture is a plant tissue culture technique involving the cultivation of an undifferentiated mass of cells (callus) on a nutrient medium under sterile laboratory conditions. | |

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| 9 | What is the difference between stem tuber and root tuber? | [K] / [CO2] |
| | Stem tubers (e.g., potato) and root tubers (e.g., sweet potato) are both underground storage organs, but they differ fundamentally in origin and anatomy: stem tubers are modified, swollen stem tissues with nodes/buds ("eyes"), while root tubers are modified, thickened roots that store nutrients | |
| 10 | How do auxins and cytokinins influence organogenesis in culturemedia? | [U] / [CO2] |
| | Auxins and cytokinins regulate plant organogenesis in culture media by acting as antagonistic, yet complementary, signaling molecules that dictate cell differentiation based on their ratio. A high auxin-to-cytokinin ratio induces root formation, a high cytokinin-to-auxin ratio induces shoot formation, and a balanced ratio promotes callus growth | |
| 11 | What is a rhizome? | [K] / [CO3] |
| | A rhizome is a modified, horizontal underground plant stem that acts as the main structural axis, sending out roots and shoots from its nodes to produce new plants | |
| 12 | What is the role of Nitrogen in rooting? | [U] / [CO3] |
| | Nitrogen is essential for root development, driving cell division, elongation, and lateral root initiation to optimize nutrient foraging . It serves as a structural component for amino acids and enzymes, facilitating energy transfer for growth while signaling the root system to adapt its structure based on available resources | |
| | | (10×1=10) |

**Part B - Short Essay - Answer any 6 questions**
**Each question carries 5 marks**

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| 13 | Explain the role of sucrose as a carbon source in micro propagation media. | [U] / [CO2] |
| | Key Roles of Sucrose in Micropropagation Media<br>❖ Carbon and Energy Source: Because in vitro plants are often mixotrophic (limited photosynthesis) or heterotrophic, they cannot produce sufficient sugars themselves. Sucrose is taken up from the media to fuel growth.<br>❖ Osmotic Regulation: Sucrose is essential for maintaining the osmotic balance of the medium, ensuring proper nutrient uptake and reducing the negative effects of in vitro stress.<br>❖ Morphogenetic Trigger: Sucrose acts as a critical signaling molecule and nutrient that promotes callus induction, somatic embryogenesis, and the proliferation of adventitious shoots.<br>❖ Optimal Nutrient: It is preferred because it is economical, easily available, and effectively hydrolyzed into glucose and fructose during autoclaving, which plant cells readily utilize.<br>❖ Effect on Growth Concentration: While 2-3% is common, studies show that different plants, can require higher concentrations (e.g., 6%) to maximize shoot elongation | |
| 14 | Describe Veneer Grafting. | [K] / [CO3] |
| | Veneer grafting is a popular, high-success propagation technique where a scion (desirable plant piece) is inserted into a shallow slit made on the side of a rootstock,<br><br>Key Characteristics & Procedure:<br>❖ Technique: A downward-inward 30-40 mm cut is made on the rootstock (at a 20 cm height) to create a "veneer" of bark/wood, with a small notch removed at the base to seat the scion.<br>❖ Scion Preparation: A long, shallow, slanted cut is made on one side of the scion (approx. 10-15 cm long, pencil-thick) to match the cut on the rootstock, ensuring cambium layers align.<br>❖ Application: The scion is inserted, fitted securely, and wrapped with grafting tape.<br>❖ Aftercare: The top of the rootstock is not immediately removed. Once the scion shows new growth (2-3 weeks), the rootstock is progressively pruned above the graft.<br>❖ Best Time: Usually done during active growth periods, such as March, May, or September, depending on the plant type. | |
| 15 | Describe the basic design layout of a micropropagation laboratory. | [U] / [CO2] |

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| | | A micropropagation laboratory is designed to facilitate the rapid clonal multiplication of plants under sterile (aseptic) conditions.<br><br>The basic layout consists of four or five distinct, specialized zones:<br>1. Washing and Cleaning Area<br>➢ Purpose: Cleaning glassware, plasticware, and preparation of materials.<br>➢ Design Features: Large, acid-resistant sinks, a drying rack or oven, and storage cabinets for clean glassware.<br>➢ Location: Usually near the entrance, separated from the sterile transfer room to avoid contaminating the clean zone<br>2. Media Preparation and Sterilization Area<br>➢ Purpose: Formulating, pH adjusting, and dispensing nutrient media, followed by sterilization.<br>➢ Equipment: Analytical balances, a pH meter, magnetic stirrer/hot plate, refrigerator/freezer for chemicals, and an autoclave or pressure cooker.<br>➢ Layout: Requires ample bench space for preparing solutions.<br>3 Aseptic Transfer Area (Inoculation Room)<br>➢ Purpose: The zone where plant tissues are inoculated into media.<br>➢ Design Features: A small, dust-free, enclosed room. It should have a positive pressure system to prevent contaminated air from entering.<br>➢ Equipment: Laminar flow hoods (horizontal or vertical), UV lights for sterilization, and tools like scalpels and tweezers.<br>4. Culture Room (Incubation Room)<br>➢ Purpose: Growing the explants under strictly controlled environmental conditions.<br>➢ Design Features: Insulated walls and specialized, adjustable shelving units with fluorescent or LED lights. Temperature and humidity (50–60%) are controlled.<br>5. Acclimatization/Transplantation Area<br>➢ Purpose: Moving in vitro plantlets to ex vitro conditions (soil) for hardening. |
| 16 | What are the advantages of vegetative propagation? [K] / [CO1] | |
| | | Vegetative propagation offers rapid, cost-effective plant reproduction, producing genetically identical clones that maintain desirable parent traits (like high fruit quality).<br><br>Key Advantages:<br>➢ Genetic Consistency: The offspring are genetically identical to the parent, ensuring desirable traits are preserved in the new generation.<br>➢ Faster Growth and Maturity: Plants reach maturity and produce flowers/fruits faster than those grown from seeds.<br>➢ Seedless Variety Cultivation: Allows propagation of plants that do not produce viable seeds or no seeds at all, such as bananas, pineapples, and seedless grapes.<br>➢ Lower Costs and Easier Methods: Often cheaper and simpler than growing plants from seeds, as it can utilize cutting, grafting, or layering.<br>➢ Disease Management: Tissue culture (micropropagation) techniques can produce disease-free plants, even from infected parent plants.<br>➢ Improved Survival Rates: New plants are derived from mature parts (roots, stems, leaves), offering better survival rates compared to fragile seedlings. |
| 17 | What are the major challenges for the construction of a glass house? [A] / [CO4] | |

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| | | The construction of a glass house presents significant engineering, environmental, and habitability challenges, largely centered on balancing structural integrity with transparency and climate control. Key obstacles include managing extreme heat gain, ensuring privacy, maintaining high-performance, and structural brittleness<br><br>Major challenges for the construction of a glass house<br>1. Environmental Control and Energy Consumption<br>2. Structural and Material Challenges<br>3. Privacy and Comfort<br>4. Maintenance and Durability<br>5. Specialized Construction Requirements |
| 18 | Explain bulbs, pseudobulbs, corms and tubers with examples. | [U] / [CO3] |
| | | Bulbs - A true bulb is a short, underground stem surrounded by fleshy, layered leaves (scales) that contain the food reserves. They have a basal plate at the bottom from which roots grow and a central bud that produces new leaves and flowers.<br>Structure: Layered, papery exterior (tunic).<br>Examples: Onion, garlic, lily.<br><br>Pseudobulbs - Pseudobulbs are thick, succulent aerial stem sections found on many orchids. Unlike the others, they are not underground; they grow above ground to store water and nutrients, which allows the plant to survive dry periods.<br>Structure: Swollen, above-ground stem.<br>Examples: Dendrobium, Oncidium (types of orchids).<br><br>Corms - A corm is a solid, swollen underground stem rather than layered leaves. Unlike bulbs, they are solid through the center and surrounded by a thin, dry, papery layer (tunic). After blooming, the old corm shrivels and new cormels form on top.<br>Structure: Solid, starchy, and rounded.<br>Examples: Gladiolus, Yam.<br><br>Tubers : Tubers are thickened, swollen parts of an underground stem or root designed to store nutrients. They differ from bulbs and corms in that they lack a basal plate and a papery tunic. They have "eyes" (nodes) from which new shoots grow.<br>Structure: Fleshy and often irregularly shaped.<br>Examples:, Caladium (stem tuber), Dahlia (root tuber), Begonia (root tuber). |
| 19 | What are the applications of humidifiers in plant propagation? | [U] / [CO4] |
| | | Humidifiers are essential tools in plant propagation, primarily used to maintain high, consistent moisture levels around cuttings and seedlings that lack established roots<br><br>Key applications of humidifiers in plant propagation<br>1. Rooting Cuttings (Stem and Leaf) Preventing Wilting and Reducing Transpiration<br>2. Seed Germination - Maintaining Medium Moisture and .Improved Success Rates<br>3. Greenhouse and Propagation Structures - Mist Chambers and Controlled Environments<br>4. 4. Tropical Plant Propagation - Replicating Natural Habitats and Preventing Leaf Damage<br>5. Tissue Culture Acclimatization - Transitioning Plantlets |
| 20 | What is the basic principle behind layering? | [K] / [CO3] |
| | | The basic principle behind layering in plant propagation is to induce adventitious root formation on a stem or branch while it remains attached to the parent plant. By keeping the new plant connected to the parent, it receives a continuous supply of water, nutrients, and carbohydrates during root development, which reduces stress and ensures a higher success rate compared to cuttings.<br><br>Principle<br>❖ Continued Nutritional Support: Because the stem is not severed until it has developed its own roots, it does |

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| | | not rely on stored energy alone, allowing it to survive better than a detached cutting.<br>❖ Wounding and Auxin Accumulation: Layering usually involves wounding (like girdling) or bending the stem. This interrupts the downward flow of carbohydrates and hormones (auxins) from the leaves, causing them to accumulate at the wounded point, which stimulates root development.<br>❖ Environmental Control: The target area is covered with a moist, rooting medium (like soil or sphagnum moss) to provide the dark, moist conditions necessary for roots to grow. |
| | | (6×5=30) |

| **Part C – Essay - Answer any 1questions<br>Each question carries 10 marks** | | |
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| 21 | Describe the unorganized methods of plant micropropagation and explain their significance in plant tissue culture | [U] / [CO3] |
| | Unorganized methods of plant micropropagation rely on the development of an undifferentiated mass of cells, known as callus, which is subsequently induced to regenerate whole plants. Unlike organized methods (such as meristem culture) that use pre-existing shoot tips, unorganized methods depend on the totipotency of callus cells or suspension cultures, offering high regeneration potential for recalcitrant plants<br><br>Unorganized micropropagation techniques include:<br><br>Callus Culture - This is the most common unorganized method, where a small piece of plant tissue (explant) is cultured on a suitable medium to produce an unorganized mass of proliferating cells.<br>▪ Mechanism: Explants are induced to dedifferentiate using specific plant growth regulators (auxins and cytokinins) to form a callus<br>▪ Application: It is useful for plant genetic engineering, as the callus cells can be used to produce genetically modified plants.<br>▪ Regeneration: The callus is later subcultured into a new medium to induce organogenesis (shoots and roots).<br>Suspension Culture - This method involves placing pieces of callus into a liquid medium that is constantly agitated.<br>▪ Mechanism: The agitation separates the callus into individual cells or small clusters of cells, which then multiply rapidly while suspended in the nutrient liquid.<br>▪ Application: Ideal for the large-scale production of specialized plant products and for creating high-volume, uniform plant material.<br>Somatic Embryogenesis - This is an in vitro technique where somatic (non-reproductive) cells form somatic embryos that resemble zygotic embryos.<br>▪ Mechanism: Cells within a callus mass are induced to form bipolar structures that look like embryos, which then grow into plantlets.<br>▪ Advantage: This method is highly efficient for mass propagation because one cell can produce multiple embryos<br>Protoplast Culture - This is a highly specialized, unorganized method where the cell walls of cells are removed to create protoplasts.<br>▪ Mechanism: The isolated protoplasts are cultured to regenerate a new cell wall, form a callus, and ultimately regenerate into a whole plant.<br>▪ Application: Used for somatic hybridization (fusing two different plant species) and genetic transformation studies. | |

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| | | Significance of Unorganized Methods:<br>❖ Extremely Rapid Mass Multiplication<br>❖ Genetic Manipulation and Engineering<br>❖ Production of Secondary Metabolites<br>❖ Somatic Embryogenesis<br>❖ Somaclonal Variation<br>❖ Protoplast Culture | |
| 22 | | Discuss the role of plant growth regulators in rooting. | [U] / [CO3] |
| | | Plant growth regulators (PGRs) play a crucial role in rooting by acting as chemical signals that manage the initiation, development, and elongation of root systems. Auxin is the master regulator, driving adventitious root formation and lateral root growth, while other hormones like cytokinins and gibberellins interact to modulate, or sometimes inhibit, root growth | |
| | | Key Roles of Plant Growth Regulators in Rooting<br>Auxins (The Master Regulator)<br>❖ Root Initiation: Induces the formation of adventitious roots, essential for rooting cuttings, through stimulation of cell division.<br>❖ Lateral Root Development: Promotes the development of lateral roots and root hairs to increase nutrient and water uptake<br>❖ .Rooting of Cuttings: Exogenous application of synthetic auxins like IBA (Indole-3-butyric acid) or NAA (Naphthaleneacetic acid) is a standard horticultural practice to promote rapid rooting.<br><br>Cytokinins:<br>❖ Regulate root meristem activity and cell division.<br>❖ Generally, high cytokinin-to-auxin ratios promote shoot development, while low ratios encourage root development.<br><br>Other Influential PGRs:<br>❖ Ethylene: Often works in conjunction with auxin to promote adventitious root formation and lateral root branching.<br>❖ Gibberellins (GAs): Generally inhibit adventitious root formation, making low levels necessary for healthy root development.<br>❖ Abscisic Acid (ABA): Manages root elongation under stress conditions, inhibiting growth during drought to conserve resources. | |
| | | | (10×1=10) |